

GMAT Focus Edition

Concise Cheat Sheet & Key-Topic Reference

Mapped to the *GMAT Official Guide 2025–2026* (GMAC)

Quantitative Reasoning	21	45 min	Problem-Solving (no calculator)
Data Insights	20	45 min	Data Sufficiency, Multi-Source Reasoning, Table Analysis, Graphics Interpretation, Two-Part Analysis (on-screen calculator)
Verbal Reasoning	23	45 min	Reading Comprehension, Critical Reasoning
Total	64	135 min	+ one optional 10-min break

Scoring: Total score 205–805 (10-pt steps); each section scored 60–90. Section-adaptive — you cannot skip, but may bookmark and edit up to 3 answers per section within the time limit. You choose the section order.

How to use this sheet: each section lists the must-know concepts, formulas, and common traps, with the relevant Official Guide chapter in the header. Use it for fast revision; drill real questions in Chapters 4, 6, and 8.

1 · Quantitative Reasoning

Official Guide Ch. 3 (Math Review) & Ch. 4 | Arithmetic + Algebra only, no calculator

Value, Order & Factors (3.1)

Divisibility	Even $\pm$ Even = Even; Even $\times$ anything = Even. Div. by 3 if digit-sum div. by 3; by 9 if digit-sum div. by 9; by 4 if last two digits div. by 4.
Primes	Exactly two factors (1 & itself). 2 is the only even prime; 1 is not prime. Know primes up to 50.
Prime factorization	Every integer >1 = unique product of primes. # of factors = product of (each exponent + 1).
GCF & LCM	GCF $\times$ LCM = product of the two numbers. LCM = highest powers; GCF = lowest powers of shared primes.
Exponents	$x^a \cdot x^b = x^{a+b}$; $x^a/x^b = x^{a-b}$; $(x^a)^b = x^{ab}$; $x^0 = 1$; $x^{-a} = 1/x^a$.
Roots	$\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b}$; $\sqrt{(x^2)} = x $. Squaring can add extraneous roots — check answers.
Absolute value	$ x \geq 0$; $ x = k \Rightarrow x = k$ or $x = -k$. $ a - b $ = distance between a and b.

Algebra, Equalities & Inequalities (3.2)

Key factoring	$a^2 - b^2 = (a+b)(a-b)$; $(a \pm b)^2 = a^2 \pm 2ab + b^2$.
Quadratics	Set =0 and factor, or $x = [-b \pm \sqrt{(b^2 - 4ac)}]/2a$. Discriminant >0 / $=0$ / $<0 \rightarrow 2/1/0$ real roots.
Systems	Substitution or elimination. 2 distinct linear eqns, 2 unknowns $\rightarrow$ one solution.
Inequalities	Flip the sign when multiplying/dividing by a negative. Never multiply by a variable of unknown sign.
Functions	$f(x)$: substitute carefully. Watch domain limits (no division by 0, no $\sqrt{\text{of negatives}}$).

Rates, Ratios & Percents (3.3)

Percent	part = percent $\times$ whole. % change = $(\text{new} - \text{old})/\text{old} \times 100$. Successive % changes multiply, not add.
Ratios	Keep units consistent; scale by a common multiplier. $a:b:c \rightarrow$ parts of total $(a+b+c)$.
Rate / Work	Distance = rate $\times$ time. Combined rate = sum of rates; time together = $1/(1/a + 1/b)$.
Average speed	= total distance / total time (NOT the average of the speeds).
Interest	Simple: $I = \text{Prt}$. Compound: $A = P(1 + r/n)^{nt}$.
Mixtures	Track the quantity of each component, not just the percentages.

Statistics, Sets, Counting & Probability (3.4)

Mean / Median	Mean = sum/count. Median = middle value (avg of two middles if even count); resists outliers.
Mode / Range	Mode = most frequent value. Range = max – min.
Std deviation	Spread about the mean; larger = more spread. Adding a constant to all values doesn't change SD.

Sets	Union count: $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B)$. Use a Venn diagram or a double-matrix for overlapping groups.
Counting	Multiplication principle: multiply choices per stage. Permutations: $nPr = n!/(n-r)!$. Combinations: $nCr = n!/[r!(n-r)!]$.
Probability	$P = \text{favorable}/\text{total}$, $0 \leq P \leq 1$. $P(\text{not } A) = 1 - P(A)$. Independent: $P(A \text{ and } B) = P(A)P(B)$. Excl.: $P(A \text{ or } B) = P(A) + P(B)$.

Quant strategy & common traps

- No calculator — look for arithmetic shortcuts; estimate/round before computing.
- Plug in smart numbers or test the answer choices when algebra gets messy.
- Answer the exact question asked (x vs. x^2 vs. the value of an expression).
- Watch units, and whether the answer is a count, fraction, or percent.
- Formula reference sheets are in Guide section 3.5.

2 · Data Insights

Official Guide Ch. 5 (DI Review) & Ch. 6 | Reason with data across formats; on-screen calculator allowed

Foundations: Data Sets, Displays & Patterns (5.1–5.3)

Data types	Quantitative (numeric) vs. categorical; discrete vs. continuous. Know what each variable measures.
Displays	Tables, bar/line/segmented charts, scatterplots, histograms. Read titles, axes, units & legend first.
Patterns	Trends, correlation (not causation!), clusters, outliers. Positive vs. negative association in scatterplots.

The Five Question Types (6.1–6.2)

Data Sufficiency	Decide if statements give ENOUGH info — you don't solve. Fixed 5 answer choices (box below).
Multi-Source Reasoning	2–3 tabs of text/tables/charts. Synthesize across sources; several questions per prompt.
Table Analysis	Sortable table; answer a set of yes/no or true/false statements. Sort the column you need.
Graphics Interpretation	Read a graph, then complete sentences via drop-down menus. Mind axes, scale & units.
Two-Part Analysis	Pick one answer in each of two columns from a shared list; the two parts are often linked.

Data Sufficiency — the 5 fixed answer choices (memorize)

A	Statement (1) ALONE is sufficient, but (2) alone is not.
B	Statement (2) ALONE is sufficient, but (1) alone is not.
C	BOTH together are sufficient, but NEITHER alone is.
D	EACH statement ALONE is sufficient.
E	Together they are still NOT sufficient.

Data Insights strategy & common traps

- Data Sufficiency: evaluate each statement ALONE first; never carry info from (1) into (2).
- Don't fully solve a DS problem — just decide whether a unique answer is possible.
- 'C trap': combining feels safe, but check whether one statement alone already suffices (answer D).
- Correlation $\neq$ causation; a single outlier can distort an average or trend.
- Read axis scales/units before choosing drop-downs; budget time — some prompts feed several questions.

3 · Verbal Reasoning

Official Guide Ch. 7 (Verbal Review) & Ch. 8 | Reading Comprehension + Critical Reasoning

Reasoning Foundations (7.1–7.3)

Analyzing passages	Find the main point, author's purpose, structure, and tone. Separate claims from evidence.
Inductive reasoning	Specific evidence → probable general conclusion. Strength depends on quality/quantity of evidence.
Deductive reasoning	General premises → conclusion that MUST follow if premises are true. Spot invalid logical leaps.
Assumption	Unstated premise the argument needs. Negate it — if the argument collapses, it was required.

Reading Comprehension (8.2, 8.4)

Approach	Map the passage: main idea, purpose of each paragraph, structure. Locate details — don't memorize them.
Question types	Main idea, supporting detail, inference, purpose/function, tone/attitude, application.
Inference	Must be supported by the text — no outside knowledge, no extreme leaps.
Wrong answers	Out of scope; too extreme/absolute; contradicts passage; half-right/half-wrong; true but irrelevant.

Critical Reasoning (8.2, 8.7)

First step	Identify the conclusion, then the premises, then the gap/assumption between them.
Strengthen	Add support / confirm the assumption / rule out an alternative cause.
Weaken	Attack the assumption / give an alternative explanation / show the evidence doesn't fit.
Assumption	Required unstated link. Test by negation.
Inference / Must be true	Only what the statements guarantee — stay within the text.
Evaluate / Flaw / Boldface	Find info that would test the argument; name the reasoning error; ID each bold part's role.

Verbal strategy & common traps

- Always pin down the conclusion before judging answer choices in Critical Reasoning.
- Extreme words (all, never, only, must) are often wrong in RC; pick the supported, moderate option.
- Beware answers that are true in the real world but not stated/implied by the passage.
- For strengthen/weaken, the right choice acts on the specific assumption — not the topic in general.
- Eliminate aggressively: cross out scope-shifts, reversals, and irrelevant comparisons.

Source: GMAT Official Guide 2025–2026, Graduate Management Admission Council (GMAC). Chapter references point to where each topic is taught in the guide. A revision aid — not a substitute for working the official practice questions.